

■ 1.7.5 Getting Pieces of Lists

<code>First[list]</code>	the first element in <i>list</i>
<code>Last[list]</code>	the last element
<code>list[[n]]</code>	the n^{th} element
<code>list[[-n]]</code>	the n^{th} element from the end
<code>list[{n₁, n₂, ...}]</code>	the list of elements $n_1, n_2, \dots$
<code>{x₁, x₂, ...} = {y₁, y₂, ...}</code>	assign each x_i to the corresponding y_i
<code>Take[list, n]</code>	the first n elements in <i>list</i>
<code>Take[list, -n]</code>	the last n elements
<code>Take[list, {m, n}]</code>	elements m through n (inclusive)
<code>Rest[list]</code>	<i>list</i> with its first element dropped
<code>Drop[list, n]</code>	<i>list</i> with its first n elements dropped
<code>Drop[list, -n]</code>	<i>list</i> with its last n elements dropped
<code>Drop[list, {m, n}]</code>	<i>list</i> with elements m through n dropped

Functions for picking out pieces of lists.

We will use this list for the examples.

```
In[1]:= t = {a,b,c,d,e,f,g}
Out[1]= {a, b, c, d, e, f, g}
```

Here is the last element of *t*.

```
In[2]:= Last[t]
Out[2]= g
```

This gives the third element.

```
In[3]:= t[[3]]
Out[3]= c
```

This gives a list of the first and fourth elements.

```
In[4]:= t[[ {1, 4} ]]
Out[4]= {a, d}
```

This sets **x** to the first element of the list, and **y** to the second one.

```
In[5]:= {x, y} = %
Out[5]= {a, d}
```

The symbol **x**, for example, now has value **a**.

```
In[6]:= x
Out[6]= a
```

This gives the first three elements.

```
In[7]:= Take[t, 3]
Out[7]= {a, b, c}
```

This gives the last three elements.

```
In[8]:= Take[t, -3]
Out[8]= {e, f, g}
```

This gives elements 2 through 5 inclusive.

```
In[9]:= Take[t, {2, 5}]
Out[9]= {b, c, d, e}
```

This gives **t** with the first element dropped.

```
In[10]:= Rest[t]
Out[10]= {b, c, d, e, f, g}
```

This gives **t** with its first three elements dropped.

```
In[11]:= Drop[t, 3]
Out[11]= {d, e, f, g}
```

This gives **t** with just its third element dropped.

```
In[12]:= Drop[t, {3, 3}]
Out[12]= {a, b, d, e, f, g}
```

$list[[i, j, \dots]]$ the element $list[[i]][[j]]\dots$
 $list[[\{i_1, i_2, \dots\}, \{j_1, j_2, \dots\}, \dots]]$
the list of elements obtained by picking out parts $i_1, i_2, \dots$ at the first level, etc.

Extracting parts of nested lists.

Here is a list of lists.

```
In[13]:= t = {{a, b, c}, {d, e, f}}
Out[13]= {{a, b, c}, {d, e, f}}
```

This picks out the first sublist.

```
In[14]:= t[[1]]
Out[14]= {a, b, c}
```

This picks out the second element in the first sublist.

```
In[15]:= t[[1, 2]]  
Out[15]= b
```

This is equivalent to `t[[1, 2]]`, but is clumsier to write.

```
In[16]:= t[[[1]][[2]]]  
Out[16]= b
```

This gives a list containing the second sublist of `t`, followed by the first one.

```
In[17]:= t[{{2, 1}}]  
Out[17]= {{d, e, f}, {a, b, c}}
```

This gives a matrix of elements of `t`.

```
In[18]:= t[{{2, 1}, {2, 3}}]  
Out[18]= {{e, f}, {b, c}}
```

Section 2.1 will show how all the functions in this section can be generalized to work not only on lists, but on any *Mathematica* expressions.

The functions in this section allow you to pick out pieces that occur at particular positions in lists. Section 2.3.5 will show how you can use functions like **Select** and **Cases** to pick out elements of lists based not on their positions, but instead on their properties.